



# Harmonizing MR Images Across 100+ Scanners: Multi-site Validation with Traveling Subjects and Real-world Protocols



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## Introduction

- Magnetic resonance (MR) images **vary across scanners, sites, and acquisition protocols**
- This variability **reduces reliability** of multi-site studies, biomarker analysis, and downstream AI models
- Harmonization aims to **reduce scanner-dependent** variability
- **Existing Challenges:**
  - Many methods **require paired** traveling-subject data
  - Existing methods, such as HACA3<sup>1</sup>, often struggle with **limited field-of-view (FOV) acquisitions**, artifacts, and scalability across scanners
  - Few methods are **validated on large** real-world datasets

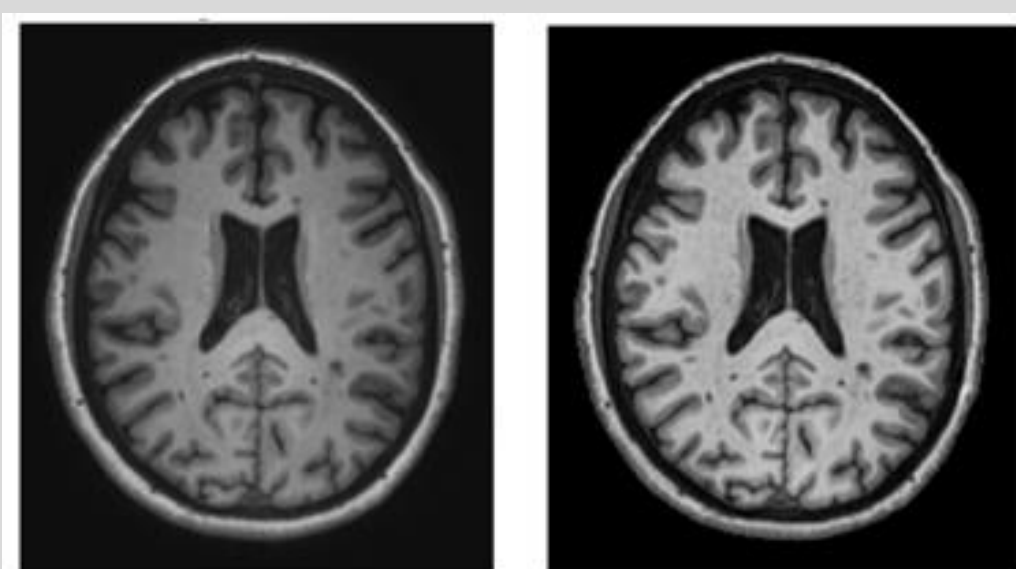


Figure 1: T1-weighted (T1w) images from the same subject acquired using different imaging protocols and scanners.

### Contributions of HACA3+

- Enhanced artifact encoder with continuous artifact severity modeling
- Foreground/background-aware attention for limited FOV recovery
- Large-scale training across 132 scanners and 996 subjects
- Validation using multiple traveling-subject datasets

## Methods

- HACA3<sup>1</sup> is an **unsupervised image harmonization algorithm** for structural MR neuroimages, which does not require paired subject data for training
- The HACA3 model is built on an **encoder-decoder** structure
- Three separate encoders **disentangle representations** of anatomy, acquisition contrast, and image quality from each input MR image
- Harmonization is achieved by **recombining anatomy and contrast encodings** to synthesize images with **desired** contrast and quality characteristics

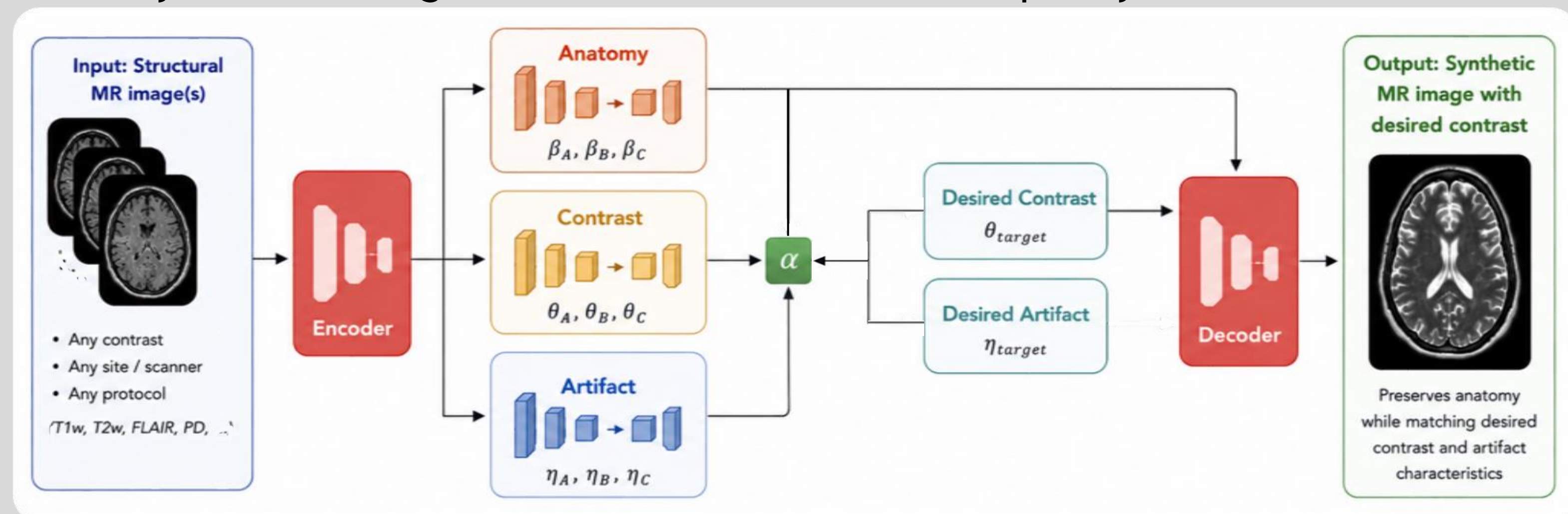


Figure 2: Overview of the HACA3 architecture, which works to disentangle anatomy, contrast, and artifacts to synthesize MR images with a desired contrast.

### Validation Datasets

- **TREAT-MS Traveling Subjects:** 15 subjects, 9 sites
- **Frequently Traveling Human Phantom (FTHP):** 1 subject, 116 sites
- **Multisite, Multiscanner, and Multisubject Acquisition for Studying Variability in Diffusion Weighted MR Imaging (MASiVar):** 5 subjects, 3 sites, 19 sessions
- **ON-Harmony:** 20 subjects, 5 sites

### HACA3+ Enhancements

#### Enhanced Artifact Encoder<sup>2</sup>

- Trained using simulated MR artifacts:
  - Noise | Ghosting
  - Bias field | Anisotropy
- Uses triplet loss with margin
- Margin is representative of the simulated artifact severity
- Learns continuous artifact representation

#### Foreground-Aware Attention

- HACA3 uses a slice-wise scalar attention
- HACA3+ uses union foreground/background masks for a spatial varying attention
- Enables limited FOV imputation
- Does NOT attempt to impute regions missing in all source images

#### Large-Scale Training

- 996 subjects, 132 scanners
- Manufacturers:
  - GE | Hitachi | Philips
  - Siemens | Toshiba
- Field strengths:
  - 1.1T | 1.5T | 3T
- Populations:
  - Healthy | Multiple sclerosis
  - Mild traumatic brain injury
- Image contrasts:
  - T1w | T2w | FLAIR | PD

## Results

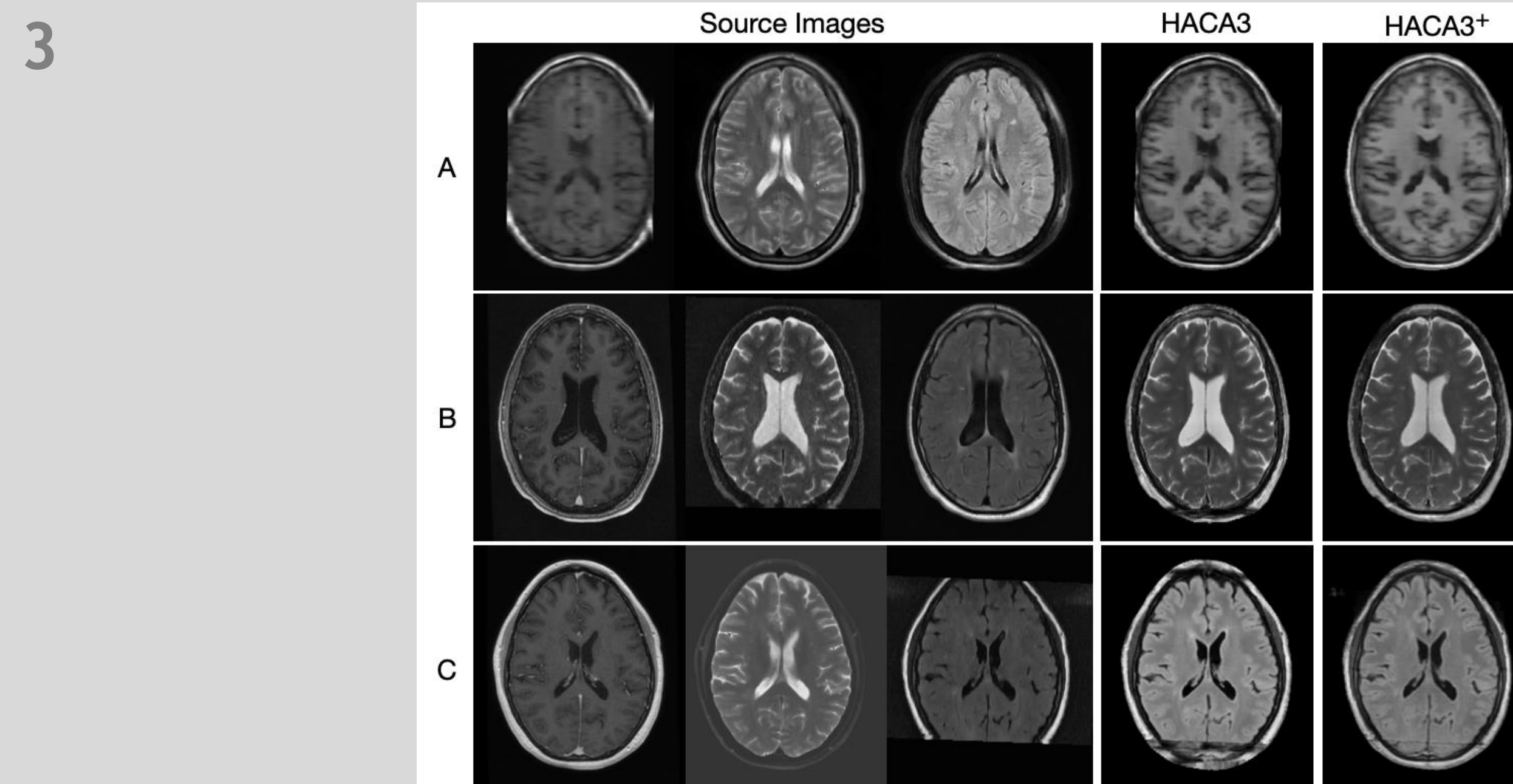


Figure 3: Qualitative results on real, clinically acquired MR images. Each row corresponds to a different person with multiple sclerosis and different limited FOV contrast.

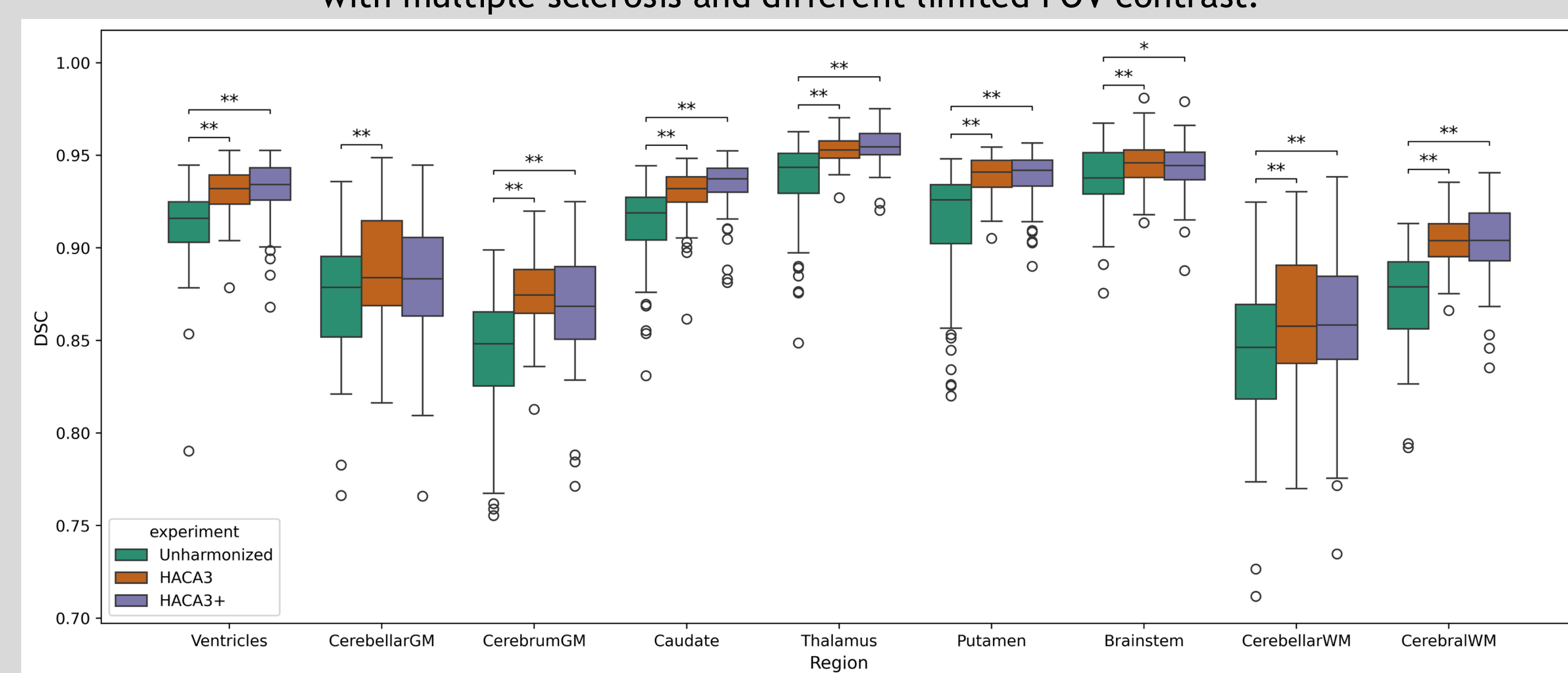


Figure 4: The DSC for each brain region computed using the segmentation result on the unharmonized, harmonized using HACA3, and harmonized using HACA3+ T1-w images from the FTHP dataset.

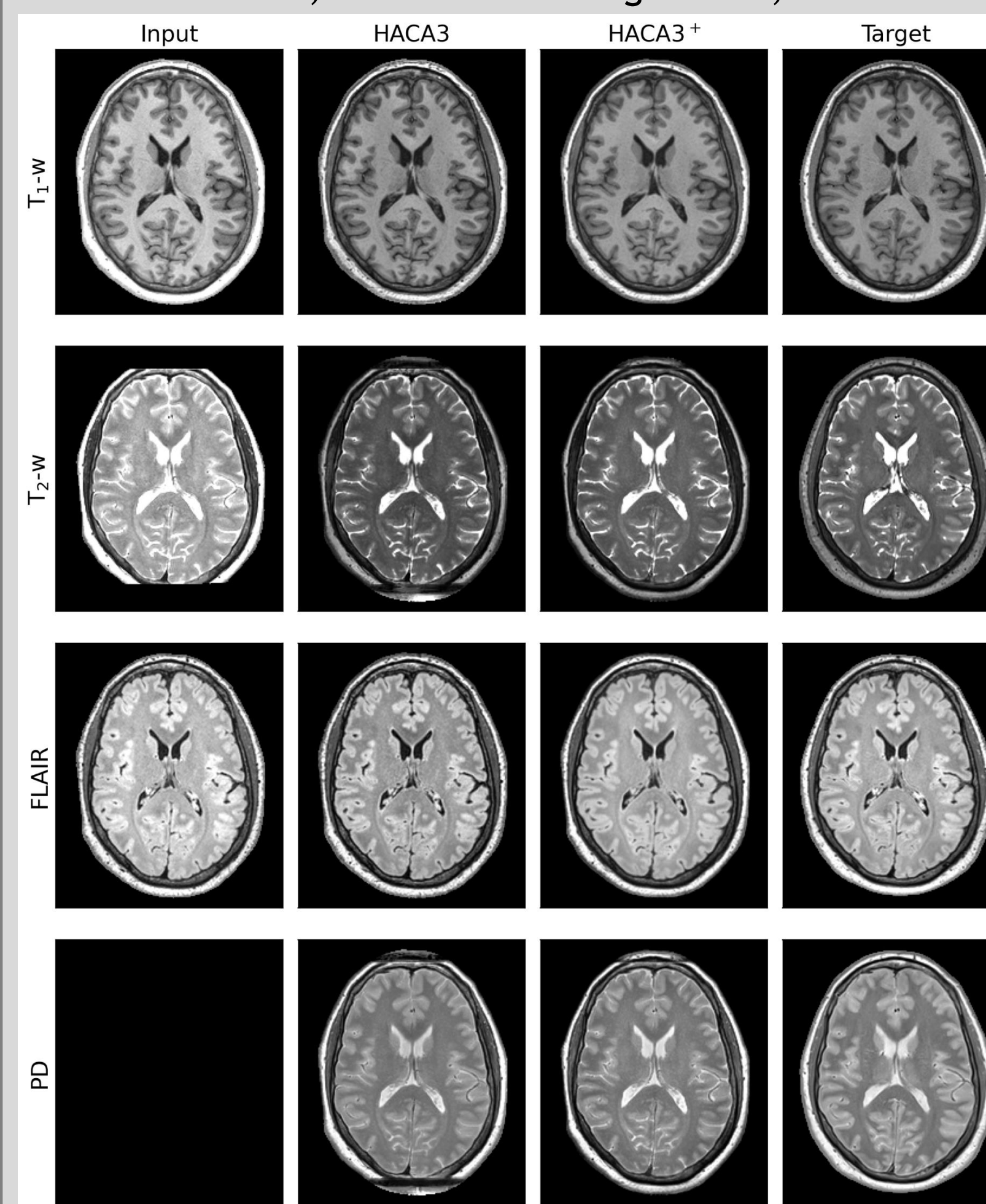


Figure 5: Inter-site harmonization results using HACA3 and HACA3+. The input PD was not included to demonstrate the imputation ability.

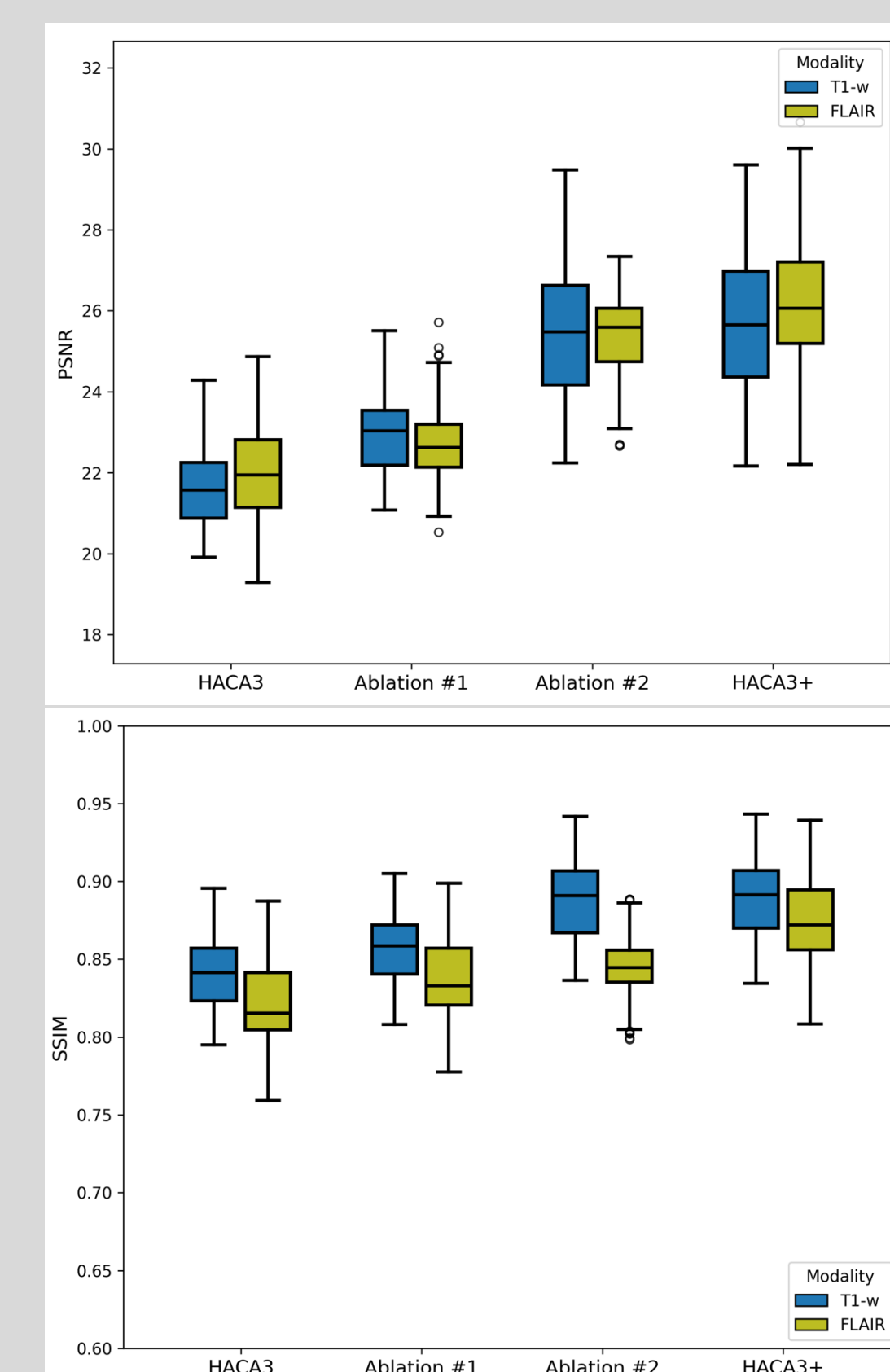


Figure 6: PSNR and SSIM from the ablation experiment using the ON-Harmony dataset. Ablation #1 is training HACA3 with enhanced attention and artifact encoder. Ablation #2 is training HACA3 with enhanced attention and large-scale training dataset.

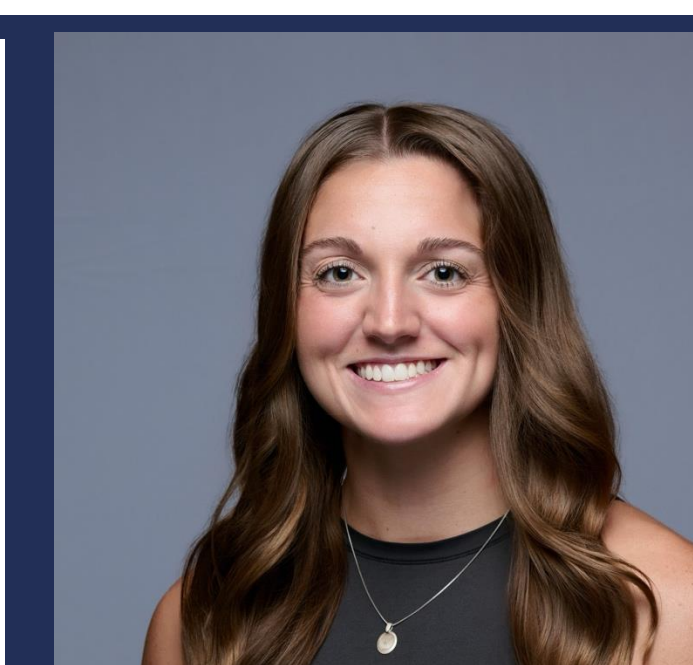
## Discussion and Conclusion

- **Key Takeaways:**
  - HACA3+ enables robust large-scale MR harmonization
  - Validated across 132 scanners from multiple vendors and field strengths
  - Demonstrated strong generalization on traveling-subject datasets
  - Reduced variability in brain segmentation DSC and volumes
- **Future Work:**
  - Extend validation to additional neurological diseases and imaging populations
  - Explore larger-capacity architectures for increasingly diverse multi-site datasets

## References

- <sup>1</sup>Lianrui Zuo et al. HACA3: A unified approach for multi-site MR image harmonization. *Computerized Medical Imaging and Graphics*, 109(102285), 2023.
- <sup>2</sup>Savannah P Hays et al. An Unsupervised Approach for Artifact Severity Scoring in Multi-Contrast MR Images. In *Medical Imaging with Deep Learning*, 2025.

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